

NLP CALLONI REPORT

Title: Simplifying the Fine Print: Balancing Readability and Information Preservation in Legal NLP

Abstract

This project investigates the capability of Large Language Models (LLMs) to perform complex text simplification in a specialized domain: legal contracts and Terms of Service (ToS). The core objective is to convert highly complex, legally binding clauses into plain language accessible to an average audience (specifically targeted at an 8th-grade reading level) while strictly preserving all original rights, obligations, and limitations. To test this, the study compared a state-of-the-art proprietary cloud-based model (Google Gemini 1.5 Flash) with an efficient, consumer-hardware-optimized open-source model (Meta Llama-3-8B, quantized to 4-bit). The pipeline evaluated the outputs across two axes: text readability (using the Flesch-Kincaid Grade Level and Gunning Fog Index) and semantic fidelity (using BERTScore and SBERT Cosine Similarity). Experimental results revealed a critical infrastructure vulnerability: the proprietary API failed entirely under rate-limiting and connection constraints in the Google Colab environment. Conversely, the local quantized Llama 3 model proved highly robust and reliable, achieving an impressive average BERTScore of 0.8616 and successfully lowering the reading complexity of the legal corpus without introducing significant hallucinations.

1. Introduction

Legal documents—such as Terms of Service, Privacy Policies, and commercial contracts—are notoriously inaccessible to the general public due to their extreme linguistic complexity, passive voice, and specialized domain jargon. This creates a significant gap in consumer protection, as users frequently agree to legally binding terms they cannot understand. Recent advances in Natural Language Processing (NLP) have positioned Large Language Models (LLMs) as potential "legal translators" capable of summarizing and simplifying these dense texts.

However, the application of LLMs to the legal domain is fraught with risk. Unlike standard text summarization, where minor omissions are tolerable, legal text simplification requires an uncompromising adherence to semantic fidelity. A model that simplifies a clause by accidentally removing a penalty or a liability constraint is not helping the user; it is generating a dangerous hallucination. This project investigates this delicate trade-off between **Readability** (making the text simpler) and **Semantic Fidelity** (ensuring no legal meaning is distorted or lost).

Specifically, this study answers two primary research questions:

1. Can LLMs effectively lower the reading level of legal texts without omitting critical constraints or hallucinating false permissions?
2. How do proprietary cloud models compare to open-source local alternatives in terms of robustness, execution stability, and domain-specific handling?

2. Dataset Selection and Preprocessing

The research proposal originally targeted the *Contract Understanding Atticus Dataset* (CUAD). However, due to streaming limitations and remote file accessibility issues during execution in the Google Colab environment, the pipeline dynamically fell back to the *LexGLUE* benchmark, specifically utilizing the "unfair_tos" (Unfair Terms of Service) subset. This dataset is ideal for the task as it contains real-world clauses extracted from consumer contracts that are often flagged as potentially infringing on consumer rights.

A total of 100 highly complex legal paragraphs were extracted from the dataset. To ensure a fair evaluation and prevent models from failing due to extreme input lengths, the data ingestion module applied strict length constraints. Only paragraphs containing between 50 and 300 words were retained. Furthermore, basic text cleaning was applied to remove excessive newlines and artifacts, and strict deduplication was enforced to ensure the 100 samples represented unique legal constraints.

3. Methodology and System Architecture

To address the research questions in a reproducible manner, a complete end-to-end Python pipeline was developed following Object-Oriented Programming (OOP) principles. The code was structured into a core `LegalSimplifier` class handling dataset loading, model inference, metric calculations, and visualization.

3.1 Models Implemented

- **Google Gemini 1.5 Flash:** Representing the closed-source State-of-the-Art (SOTA). It was accessed via the Google GenAI SDK with a temperature of 0.3 to encourage consistency and discourage creative hallucinations.
- **Meta Llama 3 8B:** Representing the open-source baseline. To run this model on Google Colab's free T4 GPU without exceeding the limited system RAM, the model was loaded using the *BitsAndBytes* library with aggressive 4-bit quantization (NF4 type) and compute data type set to float16.

3.2 Prompting Strategy

Both models were prompted with an identical, strictly structured prompt designed to prevent the loss of critical information:

"You are a legal expert translator. Simplify the following legal text for a general non-expert audience (8th-grade reading level) while strictly preserving all legal obligations, rights, limitations, and conditions. Do not omit any critical information."

3.3 Evaluation Metrics

Two distinct categories of metrics were calculated:

- **Readability Metrics:** Calculated using the `textstat` library. We computed the Flesch-Kincaid Grade Level and the Gunning Fog Index for the original text, the

Gemini output, and the Llama output. The focus was on the *Delta FKGL* (Original Score - Simplified Score), representing the net improvement in readability.

- **Semantic Preservation Metrics:** Calculated using sentence-transformers. We computed the BERTScore (F1) between the original and simplified texts to capture deep contextual overlap. Additionally, Cosine Similarity was calculated using the lightweight yet powerful all-MiniLM-L6-v2 SBERT embeddings to measure the overall vector distance between the original and simplified clauses.

4. Experimental Results and Discussion

The execution of the pipeline yielded highly unexpected results regarding model execution robustness, directly answering the second research question regarding infrastructure reliance.

4.1 The Fragility of Cloud APIs vs. Local Robustness

The most striking finding of the experiment was the complete failure of the Gemini API to process the dataset. As visible in the Colab execution logs (WARNING:tornado.access:404 POST), the batch processing of 100 texts triggered repeated connection drops or strict rate-limiting blocks. Despite the pipeline's built-in exponential backoff retry logic, the proprietary cloud tier returned ERROR_GENERATION for the samples.

On the other hand, the Meta Llama 3 8B model processed the entire batch of 100 paragraphs with a 100% success rate. Even on limited consumer-grade hardware (a single T4 GPU on Colab Free), the local, quantized model demonstrated exceptional stability and complete autonomy from internet bandwidth or external API constraints.

4.2 Quantitative Evaluation of Llama 3

Focusing on the successful Llama 3 generation:

- **Semantic Fidelity:** Llama 3 achieved a Mean BERTScore of **0.8616**. This indicates a very high degree of semantic preservation, meaning that even after aggressive rewriting, the core topics and contextual meanings of the legal clauses remained intact.
- **Readability Improvement:** The model effectively lowered the complexity of the texts. Looking at the generated boxplots and the scatter plot, the simplified versions consistently exhibited positive Readability Improvement scores.
- **Statistical Analysis:** A T-test was conducted on the semantic scores. With a highly significant p-value of 1.23e-111, the pipeline confirmed that Llama 3 was highly superior to the failing Gemini baseline in this specific environment setup.

5. Qualitative Analysis of Llama 3 Output

An inspection of the generated CSV results provides insight into *how* Llama 3 successfully navigates the trade-off between simplicity and fidelity.

- **Case 1 (Text ID 0 - High Readability Improvement):**

- *Original Text:* An extremely long sentence full of parentheses and legal jargon regarding binding agreements upon creating a Tinder account.
 - *Llama 3 Output:* The model broke the single 100-word sentence into short, digestible sentences and utilized a bulleted list to clearly outline the three components of the agreement (Terms of Use, Privacy Policy, and Additional rules). This action drastically reduced the Gunning Fog index while keeping every binding condition.
- **Case 2 (Text ID 4 - Complex Parameter Translation):**
 - *Original Text:* Mentioned granting Tinder a "worldwide , transferable , sub-licensable , royalty-free , right and license to host [...]".
 - *Llama 3 Output:* Instead of just removing these complex words, Llama 3 created a bulleted glossary explaining exactly what "Worldwide", "Transferable", and "Royalty-free" mean in plain English. This is the hallmark of an effective translation; it educated the non-expert reader rather than hiding the legal implications.

6. Conclusion

This project successfully mapped the dual evaluation plane of Readability vs. Semantic Fidelity in Legal NLP. The core findings indicate that open-source LLMs like Meta Llama 3, when quantized to 4-bit, can serve as highly competent, private, and secure legal translators on consumer-grade hardware without requiring paid cloud infrastructure. Furthermore, the total failure of the proprietary API highlighted that open-source local models are vastly superior for heavy batch-processing pipelines where network stability cannot be taken for granted. Future work will involve testing these models on longer, complete contract files to see if the semantic fidelity holds across thousands of tokens.